Type MS · Find a mean, median, mode or mean rate of change

Answer 1 of 12 · Question 5 part a.i · 2 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

5	(a)	(i)	175.9 + 178.5 + 166.3 + 167.6 ÷ 4;		1
		Clip with table	172.1;	Full marks for correct final answer Reject 172	1
		(ii)	Height increases with age (for males / females / both); Males are taller/females are shorter (at all ages); Females increase in height greater compared with males between the ages of 12 and 18; Greatest height increase between ages 12 to 14;		Max 2
		(iii)			1
			Use more students in each age group; Use a greater age range of students; Use ages with 1 year gap/13,15,17;	Allow other valid improvements Ignore repeats	1
	(b)	(i)	71 ÷ 1.46^2;		1
		Clip with table	33/33.3;	Full marks for final correct answer One mark for correct calculation with incorrect figures	1
		(ii)	student Y is obese;	ECF from b(i)	1
		CLIP (b)(i) with (b)(ii)	higher risk of diabetes/heart disease/arthritis/cancers;	Allow other medical condition linked to obesity	1

Total question 5 = 10 marks

Type MS · Find a mean, median, mode or mean rate of change

Answer 1 of 12 · Question 5 part a.i · 2 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Find a mean, median, mode or mean rate of change

Answer 2 of 12 · Question 7 part b.ii · 2 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part b.ii in this mark scheme extract

Number					
7	(a)	(i)	• allows protein to stay longer in stomach/gut (1) • gives time for full digestion (1)		2
		(ii)	• diet largely/entirely milk when young (1) • changes to solids when older (1)		2
		(iii)	• biuret test (1) • add reagent to solid (1) • blue to lilac/pale purple (1)		3
	(b)	(i)	• temperature (1) • volume of milk (1) • volume/concentration of chymosin (1)		3
		(ii)	$\frac{238 + 232 + 241 + 229}{4}$ (1) = 235 (1)		2
		(iii)	not included when calculating mean (1)		1
		(iv)	bubbles of different sizes/volume of CO_2 variable (1)		1

Total for this question = 14 marks

Type MS · Find a mean, median, mode or mean rate of change

Answer 2 of 12 · Question 7 part b.ii · 2 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Find a mean, median, mode or mean rate of change

Answer 3 of 12 · Question 5 part a · 2 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

Question number			
5 (a)	• addition of figures = 1690; • divide by 10 = 169;		2
(b)	makes it faster/decreases reaction time;		1
(c)	• caffeine has to be absorbed; • into blood; • transported to nervous system/brain;	Ignore body	3
(d)	• can't control what is in food/drink; • could contain caffeine/other factors that would affect result/to ensure caffeine is the cause;	Ignore coffee	2
(e)	185 ± 5 mm;		1 Total 9

Type MS · Find a mean, median, mode or mean rate of change

Answer 3 of 12 · Question 5 part a · 2 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Find a mean, median, mode or mean rate of change

Answer 4 of 12 · Question 6 part b.i · 2 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

<table>
<tr>
<td>6 (a) (i)</td>
<td>(cardiac muscle)

• striated/striped;
• branched;
• single/central nucleus;

(voluntary muscle)

• striated/striped;
• unbranched;
• many nuclei;

(involuntary muscle)

• non-striated/striped;
• unbranched;
• single/central nucleus;
• tapered at end;</td>
<td></td>
<td>max of 2 for each muscle

6</td>
</tr>
<tr>
<td>(ii)</td>
<td>small intestine/digestive tract/oesophagus/liver/uterus/arteries/veins/iris/sphincter muscles;</td>
<td>allow any named part of digestive tract</td>
<td>1</td>
</tr>
<tr>
<td>(b) (i)</td>
<td>1540;
(÷ 20) 77;</td>
<td>ECF incorrect addition</td>
<td>2</td>
</tr>
<tr>
<td>(ii)</td>
<td><table><tr><th>Percentage muscle mass (%)</th><th>Tally/frequency</th></tr><tr><td>70</td><td>ll 2</td></tr><tr><td>72</td><td>ll 2</td></tr><tr><td>74</td><td>llll 4</td></tr><tr><td>75</td><td>II 2</td></tr><tr><td>78</td><td>III 3</td></tr><tr><td>82</td><td>~~IIII~~ l 6</td></tr><tr><td>84</td><td>l 1</td></tr></table></td>
<td>1 mark for suitable table, 1 mark for column headings and 1 mark for correctly inputted data and tally
allow just numbers in second column
accept just percentage/% as column heading</td>
<td>3</td>
</tr>
<tr>
<td>(iii)</td>
<td>82;</td>
<td>ECF from (ii)</td>
<td>1</td>
</tr>
<tr>
<td>(c)</td>
<td>• biceps contract;
• triceps relax;
• antagonistic pairs/work antagonistically;
• lower arm/radius pulled upwards;</td>
<td></td>
<td>4</td>
</tr>
</table>

Total Marks 17

Type MS · Find a mean, median, mode or mean rate of change

Answer 4 of 12 · Question 6 part b.i · 2 marks

2024 NOV · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Find a mean, median, mode or mean rate of change

Answer 5 of 12 · Question 7 part c.ii · 2 marks

2022 JUN · Paper 01 · extract 1 of 3 · find part c.ii in this mark scheme extract

7 (a) (i)	obtain a pure product/enzyme and product not mixed/enzymes are reusable/active over wider range of temperature/pH;		1
(ii)	sucrose;		1
(iii)	any three from • add sodium alginate with distilled water; • mix (with a hand blender); • until all sodium alginate has dissolved; • add calcium chloride; • wash beads;		3
(b)	**B** insulin is released from the pancreas and travels to the liver **A** is incorrect as glucagon does not convert glucose to glycogen **C** is incorrect as glucagon is not released from the liver **D** is incorrect as insulin is not released from the liver		1]

Type MS · Find a mean, median, mode or mean rate of change

Answer 5 of 12 · Question 7 part c.ii · 2 marks

2022 JUN · Paper 01 · extract 2 of 3 · find part c.ii in this mark scheme extract

(c) (i)	• (66.65 ÷ 100) = 0.6665; • (x6) = 4.0 (million);	Allow 0.67 Allow 3.999	2
(ii)	• 38 49 51 62 65 78 82 91 96; • 65;		2
			Total 10

Type MS · Find a mean, median, mode or mean rate of change

Answer 5 of 12 · Question 7 part c.ii · 2 marks

2022 JUN · Paper 01 · extract 3 of 3 · find part c.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Find a mean, median, mode or mean rate of change

Answer 6 of 12 · Question 6 part b.iii · 1 mark

2024 NOV · Paper 01 · extract 1 of 2 · find part b.iii in this mark scheme extract

<table>
<tr><td>6 (a) (i)</td><td>(cardiac muscle)
• striated/striped;
• branched;
• single/central nucleus;
(voluntary muscle)
• striated/striped;
• unbranched;
• many nuclei;
(involuntary muscle)
• non-striated/striped;
• unbranched;
• single/central nucleus;
• tapered at end;</td><td></td><td>max of 2 for each muscle
6</td></tr>
<tr><td></td><td>(ii)</td><td>small intestine/digestive tract/oesophagus/liver/uterus/arteries/veins/iris/sphincter muscles;</td><td>allow any named part of digestive tract</td><td>1</td></tr>
<tr><td>(b)</td><td>(i)</td><td>1540;
(÷ 20) 77;</td><td>ECF incorrect addition</td><td>2</td></tr>
<tr><td></td><td>(ii)</td><td><table><tr><th>Percentage muscle mass (%)</th><th>Tally/frequency</th></tr><tr><td>70</td><td>ll 2</td></tr><tr><td>72</td><td>ll 2</td></tr><tr><td>74</td><td>llll 4</td></tr><tr><td>75</td><td>II 2</td></tr><tr><td>78</td><td>III 3</td></tr><tr><td>82</td><td>~~llll~~ l 6</td></tr><tr><td>84</td><td>l 1</td></tr></table></td><td>1 mark for suitable table, 1 mark for column headings and 1 mark for correctly inputted data and tally
allow just numbers in second column
accept just percentage/% as column heading</td><td>3</td></tr>
<tr><td></td><td>(iii)</td><td>82;</td><td>ECF from (ii)</td><td>1</td></tr>
<tr><td>(c)</td><td></td><td>• biceps contract;
• triceps relax;
• antagonistic pairs/work antagonistically;
• lower arm/radius pulled upwards;</td><td></td><td>4</td></tr>
</table>

Total Marks 17

Type MS · Find a mean, median, mode or mean rate of change

Answer 6 of 12 · Question 6 part b.iii · 1 mark

2024 NOV · Paper 01 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Find a mean, median, mode or mean rate of change

Answer 7 of 12 · Question 4 part a.ii · 1 mark

2025 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

4 (a) (i)		<table><tr><th>Conc (sugar) in au</th><th>Percentage change in mass</th></tr><tr><td>0.00</td><td>8.0</td></tr><tr><td>0.25</td><td>4.0</td></tr><tr><td>0.50</td><td>-1.0</td></tr><tr><td>1.00</td><td>-7.0</td></tr><tr><td>1.50</td><td>-10.0</td></tr></table> • two row/column table; • concentration (sugar) in au; • percentage/% change in mass; • 0.0 to 1.50/1.50-0.00;	Ignore units in the body of the table Accept AU for MP2 Ignore % symbol for MP3 mp3 unit not required for percentage in mass	4
	(ii)	0.50 (au);		1
	(iii)	$\frac{104 \times 7}{100}$; = 7.28 g; =7.3;	Allow 4/100 for 1 mark or 0.28	3
(b)		• change in mass is -1% to +4% • 5% range in mass • between 0.25 and 0.5 au; • accept any answer between 0.375- 0.5; ***an answer of o.45 scores =4 marks***	MP 1-4 are separate MPs unless the answer is 0.45	4
(c)	(i)	• dry the cylinder/ remove water/liquid; • affects mass/wrong mass/heavier/inaccurate mass/eq;	Accept weight	2
	(ii)	any two from • volume of (sugar) solution; • temperature; • surface area of cylinders; • same type of potato;	Ignore amount accept volume of liquid	2 **Total =16**

Type MS · Find a mean, median, mode or mean rate of change

Answer 7 of 12 · Question 4 part a.ii · 1 mark

2025 JUN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Find a mean, median, mode or mean rate of change

Answer 8 of 12 · Question 3 part a.i · 1 mark

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

3(a)(i)	B 3.2 - 3.5 kg;		**(1)**

Question Number	Answer	Additional guidance	Mark
3(a)(ii)	(94 / 210) x 100 ; 45%; (2)	Ecf allow correct answer to incorrect figures used in calculation Allow 44.76%/ 44.8% for correct final answer	**(2)**

Question Number	Answer	Additional guidance	Mark
3(b)	smoking appears to lower birth mass; more higher birth mass babies born to non-smokers;	ORA for both mp's	**(2)**

Question Number	Answer	Additional guidance	Mark
3(c)	• cigarette smoke contains CO; • combines with haemoglobin instead of oxygen/forms carboxyhaemoglobin; • less oxygen to fetus; • less respiration in fetus; • lower growth rate/cell division; • causes lower birth mass;		**(5 max)**

Total 10 Marks

Type MS · Find a mean, median, mode or mean rate of change

Answer 8 of 12 · Question 3 part a.i · 1 mark

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Find a mean, median, mode or mean rate of change

Answer 9 of 12 · Question 7 part b.i · 2 marks

2021 JUN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

number				
7 (a)	(i)	• temperature of incubation; • volume of milk; • composition of nutrients in agar/type of agar;		3
	(ii)	size of bacterial growth;		1
(b)	(i)	$\frac{7.1}{5} = 1.42$; $\frac{36.5}{5} = 7.3$;	One mark for each correct answer	2
	(ii)	• more bacteria in B; • at start of investigation; • because extra added to tube; • some bacteria in A/milk which grew;	Allow more growth in B/growth increases more quickly in B Allow reference to contamination e.g.from air	4
(c)		• transfer loop sterilised/heated in flame; • transfer performed quickly;	Allow reference to sterilised in context / sterile technique / sterilise equipment / named equipment	2
(d)		• repeat investigation; • using other organisms;		2

Total 14 marks

Type MS · Find a mean, median, mode or mean rate of change

Answer 9 of 12 · Question 7 part b.i · 2 marks

2021 JUN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Find a mean, median, mode or mean rate of change

Answer 10 of 12 · Question 6 part b.i · 3 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

number				
6 (a)	(i)	• both increase; • males more than females; • fluctuations; accept any two from;		3 marks
	(ii)	• layer of actively dividing cells; • can become uncontrolled; • when exposed to ultraviolet/sun light;		2 marks
(b)	(i)	17 - 48; = 31 per 100 000; $\frac{31}{25}$ = 1.2/1.24 per 100 000;	allow 16.7 - 17 and 47.8 - 48 then ECF	3 marks
	(ii)	20 per 100 000; $\frac{20 \times 7\,000\,000}{100\,000}$; = 1400;	ECF	3 marks

Total 11 marks

Type MS · Find a mean, median, mode or mean rate of change

Answer 10 of 12 · Question 6 part b.i · 3 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Find a mean, median, mode or mean rate of change

Answer 11 of 12 · Question 7 part b.i · 3 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

number				
7				
(a)	(i)	C; (pathogen) A not all bacteria cause disease B not all fungi cause disease D not all protozoa cause disease		1
	(ii)	• Ebola caused by a virus; • antibiotics only affect bacteria/not active against viruses;		2
(b)	(i)	• 700 - 180 = 520; • 520 ÷ 5 (= 104); • 100 (AU per hour);	Ecf max 2 marks if first mp incorrect but value divided by 5 to arrive at correct answer Max 1 if first answer correct but division is incorrect	3
	(ii)	• B is bactericidal; • because population decreases/kills bacteria;		2
	(iii)	• antibiotic B; • because it kills bacteria/causes population to decrease; • but symptoms may be less severe/mild/reduced; • antibiotic A leaves large numbers alive/bacterial population remains constant/only stops/reduces population growth; • these can still cause disease;		3 Total 11

Type MS · Find a mean, median, mode or mean rate of change

Answer 11 of 12 · Question 7 part b.i · 3 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Find a mean, median, mode or mean rate of change

Answer 12 of 12 · Question 1 part b.ii · 2 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

Question number					
1	(a)	(i)	• first convoluted tubule; • loop of Henlé; • second convoluted tubule;	diagram only, no labels required	3
		(ii)	44mm; 44 ÷ 650; = 0.068mm;	ecf from mp1 max 2 marks Allow 0.0676	3
		(iii)	• protein; • red/white blood cells/platelets;		2
		(iv)	glucose;		1
	(b)	(i)	Any 2 from: • same person; • same activity levels; • same volume of water; • same food eaten;		2
		(ii)	710 ÷ 4; = 178 cm^3 per hour;	Allow 1 mark if division by 4 is shown Allow 177.5	2
		(iii)	• less urine production/lower urine loss on day two; • more ADH secreted on second day; • more water reabsorbed; • more water lost by sweating;	Allow reverse argument for all marking points	3

Total 16 marks

Type MS · Find a mean, median, mode or mean rate of change

Answer 12 of 12 · Question 1 part b.ii · 2 marks

2022 JAN · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks